Week 2: Quantum Information Theory

Elnaz Hessami Pilehroon & Owen Waldron

September 30, 2026

Outline and Goals

1. Quantum states, density operators, and mixed states
2. Quantum channels and the effect of noise
3. Distinguishability
4. Entropy, information, and basic security intuition

Goal: introduce the main ideas and build intuition through worked examples.

Section 1

Quantum states, density operators, and mixed states

Density Matrix

If we have a probabilistic mixture of states, that is, a random process is used to prepare

$$\begin{cases} |\psi_1\rangle & \text{with probability} & p_1 \\ |\psi_2\rangle & \text{with probability} & p_2 \\ \vdots & \vdots & \vdots \\ |\psi_m\rangle & \text{with probability} & p_m \end{cases}$$

the density matrix is

$$\rho = p_1 |\psi_1\rangle \langle\psi_1| + p_2 |\psi_2\rangle \langle\psi_2| + ... + p_m |\psi_m\rangle \langle\psi_m| .$$

Density Matrix Properties

A matrix ρ is a density matrix if and only if
a) $\mathrm{Tr}(\rho) = 1$
b) ρ is positive semidefinite:

- There exists a matrix M such that $\rho = M^\dagger M$ or
- ρ is Hermitian: $\rho = \rho^\dagger$ and all of its eigenvalues are nonnegative real numbers or
- $\langle\psi|\,\rho\,|\psi\rangle \geq 0$ for every complex vector $|\psi\rangle$ of the appropriate dimension

Indistinguishability

Definition (Indistinguishability)

Two quantum states are said to be *indistinguishable* if, for any measurement procedure, the probability distribution of outcomes is identical between the two states.

The indistinguishability game

Challenger		**Adversary**
	$\xrightarrow{\rho_0,\ \rho_1}$	[knows both descriptions]
$b \xleftarrow{\$} \{0,1\}$		
$\tau \leftarrow \rho_b$	$\xrightarrow{\tau}$	[any measurement $\mathcal{M}$]
	$\xleftarrow{b'}$	outputs $b' \leftarrow \mathcal{M}(\tau)$

The adversary wins if $b' = b$. The states ρ_0 and ρ_1 are indistinguishable exactly when the best adversary can only win with probability $1/2$.

Exercise: Distinguishing states

Are these states distinguishable?

1. The states $|0\rangle$ and $|1\rangle$.

Exercise: Distinguishing states

Are these states distinguishable?

1. The states $|0\rangle$ and $|1\rangle$.
2. The states $|+\rangle$ and $\begin{cases} |0\rangle & \text{with prob. } 1/2 \\ |1\rangle & \text{with prob. } 1/2 \end{cases}$

Exercise: Distinguishing states

Are these states distinguishable?

1. The states $|0\rangle$ and $|1\rangle$.
2. The states $|+\rangle$ and $\begin{cases} |0\rangle & \text{with prob. } 1/2 \\ |1\rangle & \text{with prob. } 1/2 \end{cases}$
3. The states $\begin{cases} |0\rangle & \text{with prob. } 1/2 \\ |1\rangle & \text{with prob. } 1/2 \end{cases}$ and $\begin{cases} |+\rangle & \text{with prob. } 1/2 \\ |-\rangle & \text{with prob. } 1/2 \end{cases}$

Trace distance

Definition (Trace norm)

For a normal matrix M, the trace norm of M is defined as

$$\|M\|_1 = \mathrm{tr}\left(|M|\right) = \mathrm{tr}\left(\sqrt{M^* M}\right)$$

Here $|\cdot|$ and $\sqrt{\cdot}$ are applied to the eigenvalues of the spectral decomposition of M.

Trace distance

Definition (Trace norm)

For a normal matrix M, the trace norm of M is defined as

$$\|M\|_1 = \mathrm{tr}\left(|M|\right) = \mathrm{tr}\left(\sqrt{M^* M}\right)$$

Here $|\cdot|$ and $\sqrt{\cdot}$ are applied to the eigenvalues of the spectral decomposition of M.

Definition (Trace distance)

Let ρ, σ be the density operators for two quantum states. The trace distance between these states is defined as

$$\|\rho - \sigma\|_1$$

Some sources (e.g. Nielsen & Chuang) add a factor of $1/2$ here; we follow the QIC 710 course notes and omit it.

The Holevo–Helstrom Theorem

Theorem (Holevo–Helstrom)

Let ρ_0 and ρ_1 be the density matrices of two d-dimensional quantum states. If one of these two states is prepared by the flip of a fair coin, then the best distinguishing procedure succeeds with probability

$$\frac{1}{2} + \frac{1}{4} \left\| \rho_0 - \rho_1 \right\|_1$$

Exercise: Trace distance and distinguishability

Find the trace distance between:

1. The states $|0\rangle$ and $|1\rangle$.

Exercise: Trace distance and distinguishability

Find the trace distance between:

1. The states $|0\rangle$ and $|1\rangle$.
2. The states $|+\rangle$ and $\begin{cases} |0\rangle & \text{with prob. } 1/2 \\ |1\rangle & \text{with prob. } 1/2 \end{cases}$

Exercise: Trace distance and distinguishability

Find the trace distance between:

1. The states $|0\rangle$ and $|1\rangle$.
2. The states $|+\rangle$ and $\begin{cases} |0\rangle & \text{with prob. } 1/2 \\ |1\rangle & \text{with prob. } 1/2 \end{cases}$
3. The states $\begin{cases} |0\rangle & \text{with prob. } 1/2 \\ |1\rangle & \text{with prob. } 1/2 \end{cases}$ and $\begin{cases} |+\rangle & \text{with prob. } 1/2 \\ |-\rangle & \text{with prob. } 1/2 \end{cases}$

Partial Trace

If (A, B) is any pair of systems with classical state sets Σ and Γ, and we have a density matrix ρ describing a state of (A, B), the reduced states of the systems A and B are:

$$\rho_A = \mathrm{Tr}_B(\rho) = \sum_{b \in \Gamma} \left(\mathbb{I}_A \otimes \langle b|\right) \rho \left(\mathbb{I}_A \otimes |b\rangle\right)$$

$$\rho_B = \mathrm{Tr}_A(\rho) = \sum_{a \in \Sigma} \left(\langle a| \otimes \mathbb{I}_B\right) \rho \left(|a\rangle \otimes \mathbb{I}_B\right)$$

Section 2

Quantum channels and the effect of noise

Quantum Channels

A quantum channel is a map $\mathcal{E}$ on density matrices of the form

$$\mathcal{E}(\rho) = \sum_i K_i \rho K_i^\dagger, \qquad \sum_i K_i^\dagger K_i = I,$$

where the matrices K_k are called Kraus operators.

Section 3

Entropy, information, and basic security intuition

Information gain implies disturbance

Theorem (Information gain implies disturbance)

In any attempt to distinguish between two non-orthogonal quantum states, information gain is only possible at the expense of introducing disturbance to the signal.

Shannon entropy

Definition (Shannon entropy)

For a discrete random variable $X = 1, \ldots, n$ with probability distribution $p_1, \ldots, p_n$, the Shannon entropy is defined as

$$H(X) \equiv H(p_1, \ldots, p_n) \equiv -\sum_x p_x \log p_x$$

Exercise: Playing with Shannon entropy

What is the entropy associated with...

1. the toss of a fair coin?

Exercise: Playing with Shannon entropy

What is the entropy associated with...

1. the toss of a fair coin?
2. the roll of a fair die?

Exercise: Playing with Shannon entropy

What is the entropy associated with...

1. the toss of a fair coin?
2. the roll of a fair die?
3. the roll of a *Mario die* (with sides $1, 3, 3, 3, 5, 6$)? What about a *Peach die* $(0, 2, 4, 4, 4, 6)$?

Exercise: Playing with Shannon entropy

What is the entropy associated with...

1. the toss of a fair coin?
2. the roll of a fair die?
3. the roll of a *Mario die* (with sides $1, 3, 3, 3, 5, 6$)? What about a *Peach die* $(0, 2, 4, 4, 4, 6)$?
4. How do the above compare to a fair 4-sided die?

Definition (Conditional entropy and mutual information)

Let X, Y be two random variables. The joint entropy of X and Y is

$$H(X, Y) \equiv -\sum_{x,y} p(x, y) \log p(x, y)$$

Definition (Conditional entropy and mutual information)

Let X, Y be two random variables. The joint entropy of X and Y is

$$H(X, Y) \equiv -\sum_{x,y} p(x, y) \log p(x, y)$$

the entropy of X conditional on knowing Y is

$$H(X|Y) \equiv H(X, Y) - H(Y)$$

Conditional entropy & mutual information

Definition (Conditional entropy and mutual information)

Let X, Y be two random variables. The joint entropy of X and Y is

$$H(X, Y) \equiv -\sum_{x,y} p(x, y) \log p(x, y)$$

the entropy of X conditional on knowing Y is

$$H(X|Y) \equiv H(X, Y) - H(Y)$$

and the mutual information of X and Y is

$$H(X : Y) \equiv H(X) + H(Y) - H(X, Y)$$

Von Neumann entropy

Definition (Von Neumann entropy)

For a quantum state ρ, the von Neumann entropy is defined as

$$S(\rho) \equiv -\operatorname{tr}(\rho \log \rho)$$

Writing λ_x for the eigenvalues of ρ, this is equivalent to

$$S(\rho) = -\sum_x \lambda_x \log \lambda_x$$

again taking $0 \log 0 \equiv 0$.

Exercise: Computing von Neumann entropy

What is the von Neumann entropy of. . .

1. *any* pure state $|\psi\rangle$?

Exercise: Computing von Neumann entropy

What is the von Neumann entropy of. . .

1. *any* pure state $|\psi\rangle$?
2. the state $\begin{cases} |0\rangle & \text{with prob. } 1/2 \\ |1\rangle & \text{with prob. } 1/2 \end{cases}$

Exercise: Computing von Neumann entropy

What is the von Neumann entropy of. . .

1. *any* pure state $|\psi\rangle$?
2. the state $\begin{cases} |0\rangle & \text{with prob. } 1/2 \\ |1\rangle & \text{with prob. } 1/2 \end{cases}$
3. the state $\begin{cases} |0\rangle & \text{with prob. } 1/2 \\ |+\rangle & \text{with prob. } 1/2 \end{cases}$

   How does it compare to the previous answer?

Thank you!

Questions?